Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/30/2006
PI Name:	McNeil, Paul L Ph.D.		
Project Title:	Cellular and Molecular Mechanisms Mediating Rapid Resealing of Gravity-Induced Plasma Membrane Disruptions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	30912	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	02/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 9/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNA04CC44G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The major goals of the research program remain unchanged: 1) to elucidate the organelle and molecular components used by skeletal muscle to repair tears in their surface plasma membrane; 2) to identify cell and molecular events leading to muscle hypertrophy		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	Understanding of muscle disease.		

Task Progress:	We here summarize our two major areas of progress during the last year. 1) Role of Annexin A1 in Resealing Plasma membrane disruption, a common form of cell injury, is provoked by physiological levels of mechanical stress in the mammalian body. Disease can result from a failure either to prevent or repair disruptions. In Duchenne muscular dystrophy, for example, normal muscle contractions produce a pathological level of disruption injury. In limb-girdle muscular dystrophy, on the other hand, skeletal muscle cells fail to repair disruptions. A dynamic mechanism is utilized for disruption repair or resealing4. Rapid, Ca2+-triggered homotypic and exocytotic membrane fusion events erect a reparative patch and apply it to the defect site. However, the mechanism is poorly understood at the molecular level. Annexin A1 is a cytosolic protein that, when activated by micromolar Ca2+, binds to membrane phospholipids, promoting membrane aggregation and fusion. We have demonstrated during the last period of NASA funding that an annexin A1 function-blocking antibody, a small peptide competitor, and a dominant negative annexin A1 mutant protein incapable of Ca2+ binding, all inhibit resealing. Moreover, we show that, co-incident with a resealing event, a sharp increase in annexin A1 concentration occurs at the disruption site (see arrows figure below). Ca2+ entering through a disruption, we propose, locally induces annexin A1 binding to membranes and their consequent fusion. This simple mechanism can explain how a rapid, emergency fusion response can be elicited whenever and wherever needed. It does not rely on an elaborate complex of proteins pre-assembled in advance of receipt of the Ca2+ trigger, such as is present at the synapse, the classical example of a rapid, Ca2+-initiated fusion response. Instead, emergency-based fusion may utilize a protein apparatus erected in response to the Ca2+ signal and derived from components, such as annexin A1, recruited locally from the cytosol. Fusion via this recruitment mechanism is inherently flexible as regards timing and locale, explaining how an unpredictable event such as resealing can be fusion-based. This work has been submitted for publication to Nature. 2) Role of Calpain in Resealing We (in collaboration with Dr. R. Mellgren, Medical College Ohio) have also been exploring the possibility that one or more of the calpains is required for resealing. The calpains are a family of intracellular cysteine proteinases. The m- and m-calpain isozymes are ubiquitously expressed in mammalian cells, contain an essential non-catalytic small subunit, the CAPN4 gene product, and require Ca2+ for proteolytic activity. We have found that CAPN4-/- fibroblasts do not undergo Ca2+-dependent plasma membrane repair, resulting in decreased survival after membrane damage. The cytoskeletal calpain substrates talin and vimentin are exposed upon membrane damage, and are rapidly cleaved in wild-type fibroblasts, but not in CAPN4-/- cells. These studies demonstrate that typical calpains are required for rapid repair of plasma membrane damage to the cell body, and this may involve calpain-mediated cytoskeletal remodeling around the site of injury. This data has particular relevance for skeletal muscle membrane repair, as dysfunction of one of the calpains is known to result in muscular dystrophy, which can be caused by a failure to repair plasma membrane disruptions. This work will be submitted soon for publication.
Bibliography Type:	Description: (Last Updated: 06/11/2008)
Articles in Peer-reviewed Journals	McNeil A, McNeil PL. "Yolk granule tethering: a role in cell resealing and identification of several protein components." J Cell Sci. 2005 Oct 15;118(Pt 20):4701-8. Epub 2005 Sep 27. PMID: 16188936 , Oct-2005
Articles in Peer-reviewed Journals	McNeil PL, Kirchhausen T. "An emergency response team for membrane repair." Nat Rev Mol Cell Biol. 2005 Jun;6(6):499-505. Review. PMID: 15928713 , Jun-2005
Articles in Peer-reviewed Journals	Orellana-Lezcano MF, Major PW, McNeil PL, Borke JL. "Temporary loss of plasma membrane integrity in orthodontic tooth movement." Orthod Craniofac Res. 2005 May;8(2):106-13. PMID: 15888123 , May-2005